

A Reusable Ichthyometrics Framework for Monitoring Activity in High-Density Juvenile Fish Populations

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Abstract—While precision aquaculture depends on accurate phenotypic data, a “phenotyping bottleneck” persists in capturing high-resolution behavioral traits. Current methods often rely on long-term growth models or qualitative welfare markers, which fail to provide quantitative data on 3D physical activity. To address this, this study introduces a high-throughput framework that integrates deep learning object detection information with binocular stereo vision to monitor the activity of juvenile tilapia in real-time. Our framework automates non-contact fish body length estimation and reconstructs continuous 3D swimming trajectories, allowing for the precise measurement of velocity and acceleration without the distortions inherent in 2D bounding boxes associated with deep learning models. As the first application to estimate physical swimming speeds in free-roaming juveniles, this approach establishes robust circadian locomotor baselines. It serves as both an objective metric for fish vitality and an early warning system for physiological stress.

Keywords—Deep learning, Juvenile fish activity, Phenotypic trait, Stereo vision, Tilapia

I. INTRODUCTION

Aquaculture has established itself as a fundamental pillar of global food security, with selective breeding emerging as a pivotal strategy for enhancing production resilience [1]. While traditional selection programs relied heavily on the manual measurement of phenotypic traits—primarily body length and weight [2]—the advent of modern genomic tools and data-driven methodologies has enabled the development of high-yielding, disease-resistant strains with superior economic efficiency. Within this evolving framework, the accurate quantification of morphological and behavioral traits, such as biomass, linear dimensions, and locomotor performance, is indispensable for evaluating genetic gains, growth kinetics, and feed conversion efficiency.

However, the phenotypic expression of species such as Tilapia is highly sensitive to environmental variables, including water quality, temperature, and stocking density [3]. This sensitivity complicates the disentanglement of genetic effects from external environmental influences. Consequently, contemporary breeding programs increasingly face a “phenotyping bottleneck,” where the rapid expansion of genomic

data far outpaces the availability of high-resolution phenotypic measurements [4]. While genomic information represents biological potential, actualized growth trajectories are dynamic responses to environmental stressors. Static, single-point manual measurements are insufficient to capture these complex reaction norms. Therefore, providing longitudinal phenotypic data with fine temporal resolution is essential to deciphering how different genotypes manifest their growth potential under fluctuating conditions.

Addressing this gap requires the integration of genomic data with high-throughput, objective phenotyping approaches. In fish breeding research, the monitoring of juveniles (aged 1–4 months) is critical for strain improvement, as this period maximizes data stability and minimizes biological noise associated with sexual maturity. Nevertheless, large-scale acquisition of phenotypic data for juvenile fish remains technically challenging. Conventional manual capture methods induce significant physiological stress and immunosuppression [5]. Furthermore, in intensive breeding environments, precise monitoring is impeded by high stocking densities, severe inter-object occlusions, and the high metabolic rates of juveniles, which result in abrupt pose variations and non-rigid deformations.

In the domain of non-contact ichthyometrics, prevailing methodologies have established for various automated biometric estimations [6], [16], [17]. Specifically, the AIoT framework studied in [6] enables accurate non-contact measurement of fish body length, along with other environmental data, which could form the core of morphological analysis. While these methods provide high-precision data for static physical traits, the high-frequency quantification of kinematic activity remains an underdeveloped frontier. Existing methodologies in smart aquaculture often focus on macro-level system integration, such as AIoT-driven frameworks for intelligent farming [7]. While these systems excel at aggregating environmental sensor data and performing generalized 2D behavioral classification, they inherently lack the spatial depth required to compute reliable 3D swimming velocities. Other approaches focus on long-term growth simulations or qualitative welfare

indicators, such as the 3D trajectory estimations utilized in Norwegian Atlantic salmon research [8]. However, these systems are primarily optimized for large adult fish in open-ocean cages and often fail to accurately quantify the instantaneous 3D physical exertion of free-roaming juveniles in high-stocking indoor tanks.

To bridge the gap between high-speed visual detection and the requirements of precise physical measurement, this study extends established phenotyping frameworks by integrating a deep learning object detector (e.g. YOLO, You Only Look Once) with binocular stereo vision to capture dynamic behavioral data. By integrating the non-contact body length measurement from [6] with a well-defined filtering mechanism, our framework translates AI-driven perceptual capabilities into precise geometrical movement data. Unlike 2D-based activity assessments, which are often unreliable due to perspective distortions, our framework reconstructs 3D spatial coordinates to quantify real-world activity levels.

This synergistic approach enables the automated acquisition of high-fidelity locomotor data—including 3D swimming velocity and acceleration—derived directly from the spatio-temporal analysis of the identified specimens. To the best of our knowledge, this study represents the first attempt to estimate the true physical swimming speeds of free-roaming juvenile fish by building upon established ichthyometric models. By establishing a continuous circadian baseline of locomotor rates, our framework can automate health monitoring and serve as an early warning system for physiological anomalies, providing a more direct and objective metric for assessing fish vitality than existing growth-oriented or 2D-based models.

II. RELATED WORK

A. Deep Learning-based Object Detection

While two-stage deep learning-based detectors such as Region-based Convolutional Neural Networks (R-CNN) and Transformer-based models [7] offer high accuracy, their latency is prohibitive for edge devices. Thus, the one-stage object detector achieves an optimal equilibrium between speed and precision, delivering the rapid processing required for high-density, dynamic underwater monitoring [9]. For the specific task of activity level estimation, highly stable object localization is a fundamental prerequisite. While this study adopts a lightweight one-stage object detector (e.g., YOLO) to generate robust bounding boxes, the proposed framework is fundamentally detector-agnostic and can flexibly integrate any modern architecture. Instead of relying on complex pixel-level semantic segmentation, the framework directly utilizes the geometric centroids of these high confidence bounding boxes as stable spatial anchors. This strategy effectively localizes the fish in the 2D image plane, providing a reliable and computationally efficient foundation for subsequent 3D stereoscopic tracking and instantaneous velocity calculations.

B. Object Detection and Stereo Vision

To quantify the swimming kinematics and activity levels of fish, continuous temporal tracking within a 3D spatial

domain is a fundamental prerequisite. Recent advancements in deep learning, particularly the YOLO series coupled with Multi-Object Tracking (MOT) algorithms, have demonstrated robust performance in localizing and tracking multiple fast-moving targets in dynamic aquatic environments. However, 2D planar trajectories inherently lack the spatial depth necessary for calculating absolute biophysical metrics, such as instantaneous velocity. Consequently, binocular stereo vision, specifically utilizing algorithms like Semi-Global Block Matching (SGBM) [10], is widely integrated to compute disparity maps and reconstruct 3D coordinates [11].

Despite its utility, extracting stable 3D trajectories for free-roaming juvenile fish encounters unique bio-optical challenges. The body surfaces of fish fry are weakly textured, and their distal fins are semi-transparent, which frequently induces correspondence ambiguities during SGBM processing. When combined with severe inter-object occlusions in high-density tanks, these visual artifacts generate high-frequency bounding box jitters and erroneous depth jumps. In the context of kinematic analysis, these spatial discontinuities manifest as “phantom movements” in the 3D trajectory even when the fish is resting, thereby necessitating robust downstream temporal filtering to prevent the systematic overestimation of activity levels.

C. Activity Level Estimation

Computer vision-based monitoring of swimming kinematics is a vital non-invasive method for quantifying metabolic health, satiation, and environmental stress in aquaculture [12]. Although multi-object tracking (MOT) [13] is commonly applied to establish continuous swimming trajectories, raw 3D trajectory data is often compromised by stochastic noise. High-frequency detection jitter and depth estimation errors frequently induce “phantom movements” during periods of fish quiescence, leading to a significant overestimation of actual swimming intensity.

Conventional smoothing techniques, such as moving averages or Kalman filters, often fail to properly capture the distinctive “burst-and-coast” locomotor patterns of juvenile fish. These traditional methods face an inherent trade-off: excessive smoothing suppresses noise but also attenuates the high-frequency signal components associated with rapid turns and sprints, thereby obscuring biologically significant behaviors. To overcome these limitations, this study utilizes a dual-stage refinement framework integrating a statistical trimmed-mean filter with a spatial “standstill” state decision mechanism.

This standstill gate forces the instantaneous 3D velocity to zero when the displacement of the 2D bounding box falls below a predefined threshold, successfully suppressing stationary noise while preserving high-dynamic motion fidelity. Finally, to ensure biological plausibility, an empirical upper limit is established based on the critical swimming speed (U_{crit}) of Nile tilapia under optimal thermal conditions [14]. By adopting this physiological boundary condition, the framework robustly differentiates between genuine biological acceleration and systemic measurement errors.

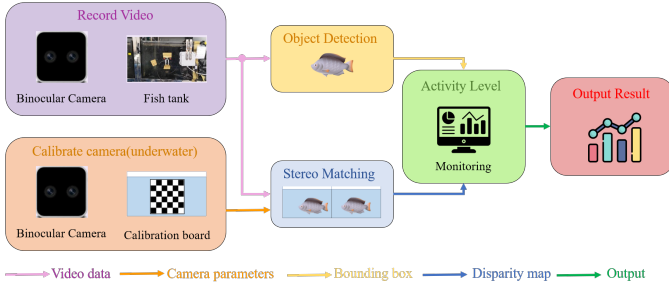


Fig. 1. Architecture and dataflow of our proposed platform.

III. METHODOLOGY

A. Architecture Overview

Fig 1 illustrates the flowchart of the proposed high-throughput phenotypic framework, which integrates computer vision with stereo geometry to achieve reliable quantification of fish activity. The workflow initiates with data acquisition using a binocular camera, where underwater calibration is applied to mitigate refractive distortion and obtain precise camera parameters. Subsequently, the video data is processed through two parallel upstream branches sharing the same input source: an object detection module that localizes the fish to generate 2D bounding boxes, and a stereo matching algorithm that computes a dense disparity map. This parallel processing ensures that visual localization and depth information are temporally synchronized for downstream information fusion. The computed level data will be collected in real time and sent to cloud for future utilization.

Leveraging these shared spatiotemporal features, the system proceeds to the behavioral estimation phase. The detected 2D bounding boxes are mapped onto the 3D disparity space, where positional and depth data are analyzed to evaluate the fish’s activity level based on reconstructed swimming trajectories. Ultimately, these behavioral metrics can be aggregated with environmental data to perform genotype-by-environment (G×E) analysis.

B. Object Detection and Stereo Vision

To acquire reliable 3D swimming trajectories, the system synchronizes 2D target localization with binocular depth estimation. First, the visual data is captured using a stereo depth camera configured for close-range measurements. To overcome the complex light refraction at the water-glass-air interface, an in-situ calibration strategy using Zhang’s algorithm [15] is performed directly within the water tank, effectively encapsulating the refractive index into the camera’s intrinsic parameters. Following stereo rectification, the disparity map is computed using the Semi-Global Block Matching (SGBM) algorithm. Concurrently, an off-the-shelf object detector is employed to localize the fish in the image plane. To prevent bounding box jitter from propagating into the downstream 3D reconstruction, a strict confidence threshold of 0.65 is applied, prioritizing spatial fidelity over marginal detection recall. Furthermore, a tracking algorithm (e.g., ByteTrack) is

integrated to associate the bounding boxes across consecutive frames, assigning a unique ID to each individual and establishing temporally continuous 2D trajectories. Rather than relying on intensive keypoint annotations, the system explicitly extracts the geometric centroid of each detected bounding box to represent the fish’s spatial location. By mapping these 2D spatial centroids onto the 3D disparity space using the calibrated focal length and baseline, the system dynamically derives the continuous 3D world coordinates (x, y, z) of each tracked subject and appends them to a spatial coordinate buffer (P_{buf}). This transformation establishes a rigorous geometric basis for subsequent kinematic analysis.

C. Activity Level Estimation

The fish activity-level analysis starts with the precise localization of fish within 2D images. For each frame, the chosen detector outputs a 2D bounding box for the fish. To convert these regional detection results into point data suitable for trajectory calculation, the system adopts a sequential processing strategy. First, the geometric centroid of each bounding box is extracted; this centroid serves as the representative position of the fish on the image plane. The 2D coordinates are then used as input for stereo vision to calculate the disparity value and transform it into 3D world coordinates $P_k = (x_k, y_k, z_k)$, based on which the velocity of each individual fish is calculated.

However, raw 3D velocities often contain jittering interferences due to depth fluctuations. To mitigate this issue, we define a “standstill” state to filter out jitters caused by object detectors. Specifically, if the inter-frame planar displacement (d_{planar}) of the bounding box centroid is below 1.5 mm and the concurrent depth variation (d_{depth}) is less than 2.0 mm, the tracked fish is considered near-static, and its instantaneous 3D velocity is strictly suppressed to zero. Such thresholding ensures that the system only collects genuine swimming maneuvers, effectively filtering out sensory jitters. Only the gated velocities are processed in the next stage. Note that this 1.5 mm physical threshold is specific to the camera optics and baseline object distance of the current setup, and scales directly with observation distance rather than biological fish size.

To eliminate detection-induced jitter, we propose a sliding-window velocity estimation algorithm. Using a non-overlapping 6-frame window, where any incomplete residual frames are discarded before restarting the next 6-frame sampling process, the local 3D swimming trajectory starting at frame i is defined as follows. Here, P_k denotes the spatial coordinate of the k -th frame.

$$P_i = \{P_k \mid k = i, i + 1, \dots, i + 5\} \quad (1)$$

here, i represents the starting frame index of the current sliding window. Second, the instantaneous velocity is calculated. Based on the aforementioned coordinate sequence, the system calculates the displacement between adjacent frames within the window. To facilitate comparisons across individuals and

studies, all velocities are normalized to body lengths per second (BL/s). The instantaneous velocity v_k between the k -th and $(k+1)$ -th frames is calculated as follows:

$$v_k = \frac{\|P_{k+1} - P_k\|}{\Delta t \cdot L} = \frac{\sqrt{(x_{k+1} - x_k)^2 + (y_{k+1} - y_k)^2 + (z_{k+1} - z_k)^2}}{\Delta t \cdot L} \quad (2)$$

where $\|\cdot\|$ denotes the Euclidean norm, representing the straight-line distance in 3D space; Δt is the time interval between frames; and L is a predefined average body length of the monitored juvenile fish. This step yields a set, denoted as V_{buf} , containing five instantaneous velocity values. Finally, the displayed speed uses averaged values because raw instantaneous velocities are susceptible to outliers caused by minor jitters in the object detection bounding boxes, and taking a direct arithmetic mean may lead to distorted results. Therefore, this study adopts trimmed-mean filtering, whereby the five instantaneous velocities are first sorted in ascending order; then, the maximum and minimum values are regarded as potential noise and discarded, and the remaining intermediate values are averaged. Thus, the final velocity, $v_{\text{disp}}(i)$, is given as

$$v_{\text{disp}}(i) = \frac{1}{3} \sum_{j=2}^4 \text{sort}(\{v_k\}_{k=i}^{i+4})_j \quad (3)$$

where $\{v_k\}_{k=i}^{i+4}$ is the set of velocities generated by Eq. (2), and $\text{sort}(\cdot)_j$ represents the order statistic, denoting the j -th value in the set after sorting. To quantify the fish's circadian rhythm over a prolonged monitoring period, this study employs a segment-wise aggregation strategy. The continuous monitoring timeline is discretized into M non-overlapped sampling intervals, e.g., 10-minute video segments.

For each sampling interval k ($k = 1, 2, \dots, M$), the proposed algorithm calculates the average activity index $\bar{V}_k$, defined as the arithmetic mean of all valid instantaneous velocities within that interval. This process compresses high-frequency kinematic data into a representative activity metric:

$$\bar{V}_k = \frac{1}{N_k} \sum_{t=1}^{N_k} v_{t,k} \quad (4)$$

where N_k is the total number of valid frames in the k -th interval, and $v_{t,k}$ denotes the filtered instantaneous velocity at frame t . Importantly, to ensure data integrity, any velocity readings exceeding the physiological burst limit, determined as 78 cm/s based on L_{max} and a 10 BL/s limit [14], are excluded from this calculation. Subsequently, a standard Circadian Activity Baseline β is constructed as an ordered sequence of these time-indexed activity metrics:

$$\beta = \{(\tau_k, b_k) \mid k = 1, 2, \dots, M\} \quad (5)$$

where $b_k = \bar{V}_k$, τ_k represents the time of day, and b_k serves as the normative activity level for that specific time block. This baseline β provides a reference profile for detecting behavioral deviations.

To detect health or environmental anomalies, we compare the new observed sequence $C = \{c_1, c_2, \dots, c_M\}$ against the baseline β using a score defined with Root Mean Square Error (RMSE), as shown in Eq. (6). As prescribed in Eq. (7), an alert is triggered if the deviation exceeds $2\sigma_{\text{base}}$, effectively distinguishing genuine anomalies from normal biological variability.

$$\text{Score} = \sqrt{\frac{1}{M} \sum_{k=1}^M (c_k - b_k)^2} \quad (6)$$

$$\text{Alarm}_{\text{trigger}} = \begin{cases} \text{True (Abnormal)}, & \text{if } \text{Score} > 2 \cdot \sigma_{\text{base}}, \\ \text{False (Normal)}, & \text{otherwise.} \end{cases} \quad (7)$$

Pseudo Code for Fish Activity Estimation

Input:

Stereo images ($I_L^{\text{raw}}, I_R^{\text{raw}}$) Camera parameters (K, D_{coeff})
Tracking ID set (IDs) Time interval (Δt) Body length (L) Burst
limit (U_{crit}) Aggregation interval (M)

Output:

Filtered speed (v_{disp}), Activity baseline (β), Alarm_{trigger}

```

1 // Phase 1: Stereo Vision Pre-processing
2 Rectify  $I_L^{\text{raw}}$  and  $I_R^{\text{raw}}$  to obtain aligned images  $I_L$  and  $I_R$ 
3 Compute disparity map  $D_{\text{map}}$  using SGBM
4
5 // Phase 2: Object Detection and 3D Mapping
6 For each tracked fish  $i$  in  $IDs$  do:
7 Extract 2D geometric centroid  $(c_{x,k}, c_{y,k})$  from the bounding box at
  frame  $k$ 
8 Project  $c_k = (c_{x,k}, c_{y,k})$  to 3D space using  $D_{\text{map}}$ ,  $K$ , and  $D_{\text{coeff}}$ 
9 Obtain  $P_k = (x_k, y_k, z_k)$ 
10 Append  $P_k$  to  $P_{\text{buf}}$  of fish  $i$ 
11
12 ...
13 // Phase 3: Activity Estimation
14 If  $\text{Size}(P_{\text{buf}}) < 6$ :
15 Continue accumulating data
16 Else:
17  $V_{\text{buf}} \leftarrow$  Empty array
18
19 For  $j = k - 4$  to  $k$ :
20 // Standstill state gatekeeper
21  $d_{\text{planar}} \leftarrow \text{Distance}(c_j, c_{j-1})$ 
22  $d_{\text{depth}} \leftarrow |z_j - z_{j-1}|$ 
23
24 If  $d_{\text{planar}} < 1.5$  mm AND  $d_{\text{depth}} < 2.0$  mm:
25  $v_j \leftarrow 0$ 
26 Else:
27  $distance \leftarrow \text{Distance}(P_j, P_{j-1})$ 
28  $v_j \leftarrow distance / (\Delta t \cdot L)$ 
29
30 Append  $v_j$  to  $V_{\text{buf}}$ 
31
32 Sort  $V_{\text{buf}}$  in ascending order
33 Remove the maximum and minimum values
34  $v_{\text{disp}}(k) \leftarrow \text{Mean}(\text{remaining three values})$ 
35
36 If  $v_{\text{disp}}(k) > U_{\text{crit}}$ :
37 Discard  $v_{\text{disp}}(k)$ 
38 ...
39
40 // Phase 4: Circadian Rhythm Aggregation and Anomaly Detection
41 For each sampling interval  $k = 1$  to  $M$  do:
42  $\bar{V} * k \leftarrow \text{Mean}(\text{all valid } v * \text{disp in interval } k)$ 

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43 Append  $\bar{V}_k$  to current observation sequence  $C$ 
44
45  $Score \leftarrow \text{RMSE}(C, \beta)$ 
46  $\sigma_{\text{base}} \leftarrow \text{StandardDeviation}(\beta)$ 
47
48 If  $Score > 2 \cdot \sigma_{\text{base}}$ :
49 Alarm * trigger  $\leftarrow$  True
50 Else:
51 Alarm * trigger  $\leftarrow$  False

```

IV. EXPERIMENTAL RESULTS

Hardware: CPU/i9-9900 with one NVIDIA GPU/RTX-2080Ti and RAM/32G. Software: Windows 11, python 3.9.21, Cuda 11.8, cuDNN 8.7.0. YOLOv11s under PyTorch 2.0.1.

A. Activity Level Estimation

While the proposed system reconstructs continuous 3D trajectories, this study focuses on kinematic vitality rather than spatial path visualization. Therefore, fish activity was evaluated using instantaneous swimming speeds derived from the reconstructed trajectories. Eight video clips were used, where the average fish length L is 6.5cm in a fish tank containing 22 tilapia for activity estimation.

Due to the strict confidence threshold and frequent inter-object occlusions in the high-density tank, valid bounding boxes were not obtained from every frame. Across the video dataset, approximately 70% of the recorded frames yielded valid detections and were used for subsequent kinematic analysis. The speed between consecutive valid detections was first computed as the unfiltered single-point baseline. These detections were not necessarily consecutive video frames. For example, if a fish was detected in frames 1, 3, 4, 5, and 7, the corresponding speed samples were calculated between frames 1–3, 3–4, 4–5, and 5–7.

To evaluate the suppression of abnormal speed estimates, the single-point baseline was compared with two 5-point trajectory post-processing strategies. In this context, each “point” denotes an instantaneous velocity sample between two adjacent valid detections, and the total number of evaluated speed samples is denoted as N_{spd} .

The 5-point filtering strategies process instantaneous velocities in discrete batches. Each averaged velocity requires five instantaneous speed samples, which are derived from at least six valid detections. Therefore, the number of evaluated samples N_{spd} for the 5-point methods is approximately one-fifth of the single-point baseline, since incomplete residual samples at the end of each tracking sequence are discarded.

TABLE I
ABNORMAL SPEED SAMPLES USING DIFFERENT TRAJECTORY
POST-PROCESSING STRATEGIES

Method	N_{spd} (total points)	N_{abn}	Percent (%)
Single-point	417,533	8,684	2.08
5-point SMA	80,443	957	1.19
5-point trimmed mean	80,443	723	0.90

Two post-processing strategies were evaluated: 5-point Simple Moving Average (SMA), which directly averages the five speed samples, and 5-point trimmed mean, which discards the maximum and minimum values before averaging the remaining three samples. Abnormal samples were identified based on visually discontinuous “phantom displacements” or velocities exceeding the physiological upper bound $U_{\text{crit}} = 78$ cm/s. As shown in Table I, the abnormal ratio decreased from 2.08% in the unfiltered single-point baseline to 1.19% with 5-point SMA and further to 0.90% with the 5-point trimmed mean.

These results indicate that temporal averaging can suppress high-frequency detection noise, while the trimmed-mean strategy further reduces residual extreme outliers by discarding the largest and smallest values in each window. Overall, the 5-point trimmed mean achieved the lowest abnormal ratio and provided the most stable kinematic estimation among the three strategies.

The trimmed-mean filter combined with the standstill gate effectively suppressed stereoscopic high-frequency noise. As shown in Fig. 2(b), the filtered speed distribution peaked near 0 BL/s, indicating that most stationary or gentle cruising states were correctly retained. The full-range distribution in Fig. 2(a) further shows that the probability density rapidly decreased beyond 3 BL/s, while still preserving rare burst-swimming events up to approximately 7 BL/s. The mean routine speed was 1.21 BL/s.

To evaluate long-term behavioral monitoring, a 24-hour activity profile was constructed, as shown in Fig. 3. The interval-averaged swimming intensity and the moving-average trend revealed a clear diurnal pattern: activity was higher

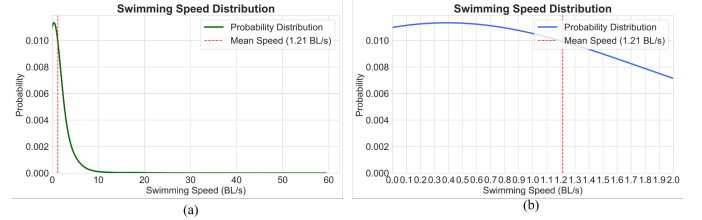


Fig. 2. Probability density of swimming speed of juvenile tilapia: (a) full-range distribution. (b) zoom-in of the 0-2 BL/s range, with the mean speed (1.21 BL/s).

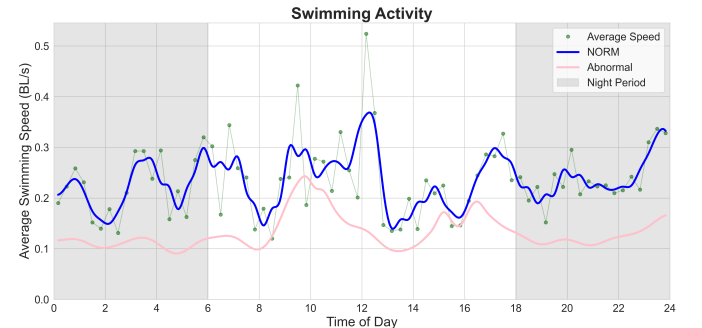


Fig. 3. Circadian activity levels derived from swimming speed, useful for monitoring the health status of juvenile fish.

and more variable during the daytime, while remaining low and stable at night. This circadian activity baseline provides a normative reference for detecting behavioral deviations. By comparing new observations with this baseline using the RMSE-based metric defined in the methodology, the proposed framework can support quantitative anomaly detection, such as hypoactivity caused by disease or hyperactivity caused by environmental stress.

To further distinguish between intensity-level anomalies, such as lethargy, and pattern-level anomalies, such as circadian disruption, we introduce Spearman's rank correlation coefficient ρ as a diagnostic metric. Unlike Pearson's correlation, which assumes linearity, Spearman's method evaluates the monotonic relationship based on rank order, making it more robust to non-linear biological variations:

$$\rho = 1 - \frac{6 \sum_{i=1}^M d_i^2}{M(M^2 - 1)} \quad (8)$$

where $d_i = \text{rank}(x_i) - \text{rank}(y_i)$ denotes the rank difference between the baseline trend value x_i and the observed activity value y_i at time step i , and M is the total number of sampling points.

Unlike Dynamic Time Warping (DTW) [16], which may align abnormal nocturnal spikes with diurnal peaks, Spearman's ρ preserves strict time-point correspondence and is therefore suitable for detecting circadian shifts. Physiologically, $\rho > 0.8$ indicates a preserved rhythm with altered intensity, such as fatigue; $\rho \approx 0$ suggests circadian disruption or random activity; and $\rho \approx -1$ implies a severely inverted activity pattern. Combined with RMSE, this metric enables a more detailed behavioral diagnosis.

Overall, the proposed framework provides multi-scale activity analysis, ranging from short-term burst swimming, to mid-term activity fluctuations, and finally to long-term circadian trends. This multi-scale capability provides a practical foundation for precision management in smart aquaculture.

V. CONCLUSION

The primary contribution of this work is the development of a scalable, non-contact framework for 3D kinematic tracking and activity-level estimation in intensive aquaculture. By extracting 2D geometric centroids from bounding boxes and mapping them into stereoscopic 3D space, the proposed system reconstructs continuous swimming trajectories without invasive measurements or labor-intensive keypoint annotations. To suppress depth-induced "phantom movements," a dual-stage kinematic filtering scheme, consisting of a standstill gatekeeper and a trimmed-mean filter, was introduced. This strategy reduced the abnormal velocity ratio from 2.08% to 0.90% while preserving burst-and-coast locomotor behaviors. By aggregating the filtered kinematic data under the physiological boundary condition U_{crit} , the framework further established a 24-hour circadian activity baseline for quantitative behavioral anomaly detection.

The proposed framework provides operational flexibility, spatial fidelity, and annotation efficiency, making it suitable

for long-term population-level monitoring in high-density aquaculture environments. Future work will integrate the established 3D kinematic information with phenotypic and environmental datasets, as shown in Fig. 1, to support smart breeding applications. These multimodal datasets can also be used to fine-tune a Large Language Model (LLM) and build an AI agent for automated health reporting, natural-language querying of physiological stress, and genotype-by-environment (G×E) analysis.

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